🎓 TECHBEAT26 — Complete Student Guide

> ***COEP AI & DS Dept. — AI Model Building Challenge 2026***
> *Script: `v2.py` | Language: Python*

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1. 🤔 What is this Script?

`v2.py` is the **official contest setup script** for TECHBEAT26. It does the following automatically:

🎨 Shows animated welcome screen	Greeting based on time of day
📋 Displays contest rules	7 rules you must follow
🏆 Explains the 3 rounds	R1.1, R1.2, and R2
🐍 Creates a virtual environment	Named `TECHBEATS26`
📦 Installs required libraries	numpy, pandas, matplotlib, scikit-learn
🚫 Removes banned libraries	tensorflow, keras, torch, etc.
📝 Generates your starter files	`model.py` and `data.py`

You run this ONCE at the start of the contest. After that, you only work on `model.py`.

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2. ✅ Prerequisites

Before you begin, make sure you have:

Python 3.8+ installed	Open terminal → type `python --version`
VS Code installed	Search "Visual Studio Code" in Start Menu
The script file `v2.py`	Should be given by the organizers

Installing Python (if not installed)

1. Go to python.org/downloads
2. Download the latest Python 3 for Windows
3. ⚠️ **IMPORTANT:** During installation, check ✅ **"Add Python to PATH"**
4. Click "Install Now"
5. Verify by opening Command Prompt and typing:

```
python --version
```

You should see something like `Python 3.12.x`

Installing VS Code (if not installed)

1. Go to code.visualstudio.com
2. Download the Windows installer
3. Install with default settings
4. **Recommended:** Install the **Python extension** (see next section)

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3. 🛠️ Setting Up VS Code

Step 1: Install the Python Extension

1. Open VS Code
2. Click the **Extensions** icon on the left sidebar (it looks like 4 squares)

• Or press `Ctrl + Shift + X`

3. Search for **"Python"**
4. Click **Install** on the one by **Microsoft** (it will be the first result)

Step 2: Choose Your Python Interpreter

1. Press `Ctrl + Shift + P` (opens the Command Palette)
2. Type **"Python: Select Interpreter"**
3. Select your installed Python version (e.g., `Python 3.12.x`)

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4. 📂 Opening the Project Folder

Method 1: From VS Code

1. Open VS Code
2. Go to **File** → **Open Folder**
3. Navigate to the folder where `v2.py` is saved
4. Click **Select Folder**

Method 2: From File Explorer

1. Navigate to the folder containing `v2.py` in File Explorer
2. Right-click in the folder
3. Select **"Open with Code"** (if available)

Method 3: From Terminal

```
cd C:\path\to\your\folder
code .
```

> ⚠ **IMPORTANT**
> **Always open the FOLDER, not just the file. This ensures the terminal starts in the correct directory.**

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5. 💻 Understanding the Terminal

The terminal is a text-based window where you type commands to run programs.

Opening the Terminal in VS Code

• Press `` `Ctrl + ` (backtick key, usually below Esc`) ``
• Or go to **Terminal** → **New Terminal** from the menu bar

Types of Terminal

PowerShell	`PS C:\..>`	Default on Windows, works fine
Command Prompt	`C:\..>`	Also works

Essential Terminal Commands

`cd folder_name`	Go inside a folder	`cd Desktop`
`cd ..`	Go back one folder	`cd ..`
`dir`	List files in current folder	`dir`
`cls`	Clear the screen	`cls`
`python file.py`	Run a Python script	`python v2.py`

> ⚠ **TIP**
> You can type the **first few letters** of a file name and press **Tab** to auto-complete it.

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6. 🚀 Running the Script

Step-by-Step

1. **Open VS Code** with the project folder (see Section 4)
2. **Open the Terminal** (press `` `Ctrl + `` ``)

3. **Verify you're in the right folder.** The terminal should show the path where `v2.py` is located. Type:

```
dir
```

You should see `v2.py` in the list.

4. **Run the script:**

```
python v2.py
```

5. **Follow the on-screen instructions** — just press **Enter** when prompted.

> ⚠ **CAUTION**
> **Do NOT close the terminal while the script is running. It needs time to create the virtual environment and install packages.**

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7. 📺 What Happens When You Run It

The script goes through several animated screens:

Screen 1 — Greeting 👋

• Shows a time-based greeting (Good Morning / Afternoon / Evening / Night)
• Displays system info (Python version, OS, time)
• **Action:** Press `Enter` to continue

Screen 2 — Contest Rules 📋

• Shows the TECHBEATS banner animation
• Displays 7 rules:
1. Individual participation only
2. Use Python only
3. No copying code
4. Don't modify the dataset
5. Code must run successfully
6. Be ready for viva
7. Rule violations → disqualification
• **Action:** Press `Enter` to continue

Screen 3 — Round Structure 🏆

• Shows all 3 rounds:

R1.1	Data Cleaning	Clean a noisy dataset
R1.2	sklearn Linear Regression	Use scikit-learn
R2	Manual Linear Regression	⚠ sklearn NOT allowed
R3	Viva	Oral explanation of your code

• **Action:** Press `Enter` to accept rules and begin setup

Setup Phase ⚙ (Automatic — just watch!)

• **Step 1/5:** Creates virtual environment (`TECHBEATS26`)
• **Step 2/5:** Upgrades pip
• **Step 3/5:** Removes restricted libraries (tensorflow, keras, torch, etc.)
• **Step 4/5:** Installs required packages (numpy, pandas, matplotlib, scikit-learn)

Round Selection 🎯

• You choose which round to attempt: **1**, **2**, or **3**
• Confirm with **Y**

Step 5/5 — Files Created 📝

• `model.py` — **Your working file** (write your code here)
• `data.py` — **Data file** (DO NOT EDIT THIS!)

Final Screen ✅

• Shows a summary of everything that was set up
• Displays "GOOD LUCK!" animation
• Activates the virtual environment automatically

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8. After Setup — Writing Your Code

What files were created?

`model.py`	Your solution goes here	✅ YES — write your code here
`data.py`	Contains the dataset & grading logic	❌ NO — do not touch this!

Opening `model.py`

1. In VS Code, click on `model.py` in the **Explorer panel** (left sidebar)
2. Or press `Ctrl + P` and type `model.py`

What's inside `model.py`?

Depending on the round you chose:

Round 1.1 (Data Cleaning):

```
import data
RAW = data.RAW # The noisy dataset
cleaned = [] # Put your cleaned data here

# YOUR CODE HERE — clean the data!

data.graph(cleaned) # DO NOT REMOVE — this grades your work
```

Round 1.2 (sklearn Linear Regression):

```
import data
import numpy as np
from sklearn.linear_model import LinearRegression
m, c = None, None # Store slope (m) and intercept (c)

# YOUR CODE HERE — train a model!

data.graph(m, c) # DO NOT REMOVE — this grades your work
```

Round 2 (Manual Linear Regression):

```
import data
m, c = None, None
X_VALUES = data.X_VALUES
Y_VALUES = data.Y_VALUES

# YOUR CODE HERE — calculate m and c WITHOUT sklearn!

data.graph(m, c) # DO NOT REMOVE — this grades your work
```

> ⚠ **WARNING**
> **Never delete the `data.graph(...)` line at the bottom! This is what checks and grades your solution.**

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9. ▶ Running Your Solution

After writing your code in `model.py`:

Method 1: From Terminal

```
python model.py
```

Method 2: Using the Test Command (Recommended)

```
python v2.py --test
```

This does 3 things automatically:

1. ✅ Checks if `data.py` is untampered
2. 🔄 Restores `data.py` if it was modified
3. ▶ Runs your `model.py`

Other Useful Flags

`python v2.py --guard`	Check if `data.py` is intact
`python v2.py --restore`	Restore `data.py` to original if tampered
`python v2.py --test`	Guard + Restore + Run model.py

Understanding the Output

When you run your code, you'll see a **results panel**:

- ✔ **PERFECT** — Your solution is correct! 🎉
- ≈ **ACCEPTABLE** — Close, but needs improvement
- ✘ **INCORRECT** — Check your logic
- 🚨 **CHEAT DETECTED** — You used a banned library in Round 2

A **matplotlib graph** will also pop up showing your results visually.

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10. 📋 Useful Commands Cheat Sheet

Run the setup script	`python v2.py`
Run your solution	`python model.py`
Test with integrity check	`python v2.py --test`
Check if data.py is intact	`python v2.py --guard`
Restore data.py	`python v2.py --restore`
Check Python version	`python --version`
List files in folder	`dir`
Clear terminal	`cls`
Open new terminal	`Ctrl + ```
Open Command Palette	`Ctrl + Shift + P`
Quick file search	`Ctrl + P`
Save file	`Ctrl + S`
Undo	`Ctrl + Z`
Run selection in terminal	Select code → `Shift + Enter`

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11. 🔧 Troubleshooting

❌ `'python' is not recognized as an internal or external command`

Problem: Python is not in your system PATH.

Fix:

1. Reinstall Python from python.org
2. **Check** ✅ **"Add Python to PATH"** during installation
3. Restart VS Code after installing

Alternative: Try using `python3` instead of `python`:

```
python3 v2.py
```

❌ `ModuleNotFoundError: No module named 'venv'`

Fix: Install the full Python distribution (not a minimal one). Re-download from python.org.

❌ Terminal shows wrong directory

Fix: Make sure you opened the **folder** in VS Code, not just the file. Go to **File → Open Folder** and select the folder containing `v2.py`.

❌ `PermissionError` or `Access Denied`

Fix: Try running VS Code as Administrator:

1. Right-click VS Code in Start Menu
2. Select "Run as administrator"

❌ Weird characters showing (like `[92m` instead of green text)

Problem: Your terminal doesn't support ANSI colors.

Fix: Use the **VS Code integrated terminal** instead of the old Command Prompt. The VS Code terminal handles colors correctly.

❌ `data.py has been modified or corrupted`

Fix: Run the restore command:

```
python v2.py --restore
```

❌ Graph window doesn't appear

Fix: Make sure `matplotlib` is installed. Run:

```
pip install matplotlib
```

Or re-run the setup script: `python v2.py`

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12. 🔍 Code Explanation (What Each Section of `v2.py` Does)

For students who want to understand the script's inner workings:

High-Level Architecture

```
v2.py
├── CONFIG → Virtual env name, allowed/banned packages
├── ANSI STYLES (class S) → Color codes for terminal animations
├── TEXT EFFECTS → typewrite(), matrix_drop(), gradient_bar()
├── BOXES & PANELS → fancy_box(), info_panel(), step_ok/fail()
├── ASCII ART → TECHBEATS banner, greeting text art
├── SCREENS → screen_greeting(), screen_rules(), screen_rounds()
├── SETUP LOGIC → venv creation, pip install/uninstall
├── FILE TEMPLATES → model.py & data.py content for each round
├── INTEGRITY GUARD → SHA-256 hash check on data.py
└── MAIN → Orchestrates everything in sequence
```

Key Components Explained

Config	30–47	Defines the venv name, packages to remove, and packages to install
Class S	54–76	ANSI escape codes for colored/styled terminal output
Text Effects	131–228	Fancy animations: typewrite, matrix rain, progress bars, spinners
Screens	463–560	The 3 interactive screens (greeting, rules, rounds)
Backup	566–585	Moves existing `model.py`/`data.py` to a `previous_attempts/` folder
Round Selection	592–611	Menu for choosing Round 1.1, 1.2, or 2
File Templates	618–1101	The actual Python code that gets written into `model.py` and `data.py`

Integrity Guard	1167–1226	Uses SHA-256 hashing to detect if a student tampered with `data.py`
Main Flow	1232–1359	Command-line argument parsing + the full setup sequence

Anti-Cheat System

The script has a built-in anti-cheat mechanism:

• **data.py** contains a unique seed generated from your computer's MAC address + hostname

• This means every student gets a **unique dataset** — you can't copy answers from others

• In **Round 2**, the `data.py` file secretly scans your `model.py` for banned terms like `sklearn`, `polyfit`, `scipy`, etc.

• If detected → 🚨 **CHEAT DETECTED** is shown instead of your grade

Hidden Feature: Skip Setup

If you've already run the setup before and just want to switch rounds, you can type **`s`** (lowercase) at any "Press ENTER" prompt to skip the setup and go directly to round selection.

🏁 Quick Start Summary

```
1. Open VS Code
2. Open the folder containing v2.py (File → Open Folder)
3. Open Terminal (Ctrl + `)
4. Run: python v2.py
5. Follow the screens → Press Enter → Pick your round
6. Open model.py → Write your solution
7. Run: python model.py
8. Check your grade → Fix if needed → Repeat step 7
9. Call the organizer when you're done!
```

> ***Good luck!*** 🍀 ***May the best coder win!***

>

> *— TECHBEAT26 Organizing Team*